

GIAC

12th
FUTURE
ARCHITECTURE
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云原生架构在 AI 时代的优化： 智能化流量管理 在 multi-agent 场景中的应用

msup[®]

ARCHNOTES
架构
可用
架构



个人简介

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- CNCF Ambassador
- Microsoft MVP
- Kubernetes Ingress-Nginx maintainer
- 『k8s 生态周报』维护者





内容概述

在 AI 时代下，流量规模和特征都有其特殊性。

技术发展也需要进行相应的迭代才能更好的为业务和企业提供价值。

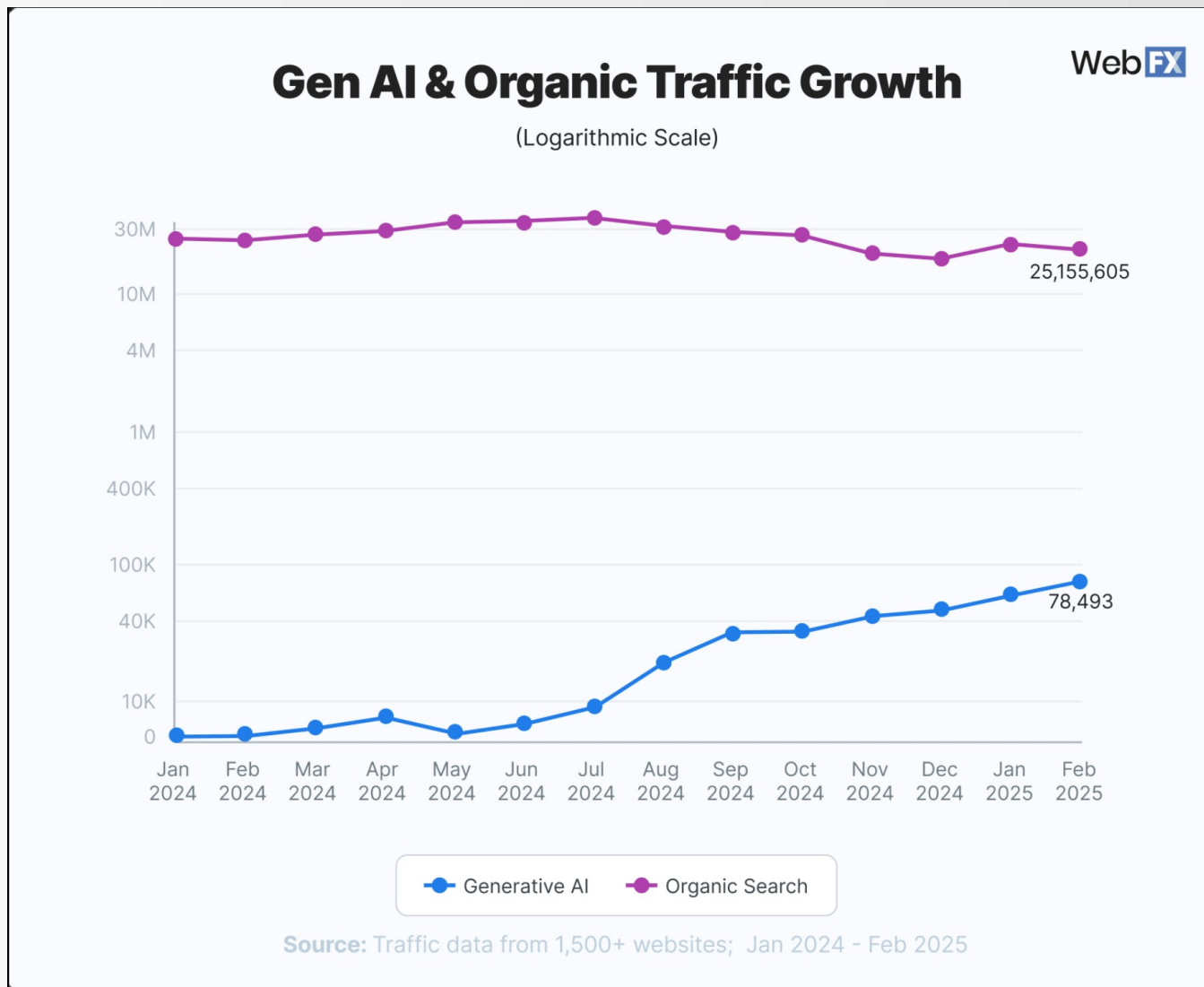


目录

- AI 时代的挑战
- Kong 如何通过 AI Gateway 及 Gateway 应对挑战
- 未来展望



生成式 AI 流量激增





什么是 MAS （ Multi-Agent System ）

- 多个 Agent 共同协作
- 去中心化 （ Decentralization ）
- 局部视图 （ Local views ）
- 自主性 （ Autonomy ）

An introduction to Multi-Agent System

https://uranos.ch/research/references/Wooldridge_2001/TLTK.pdf



AI Agent vs Multi-Agent

Single Agent

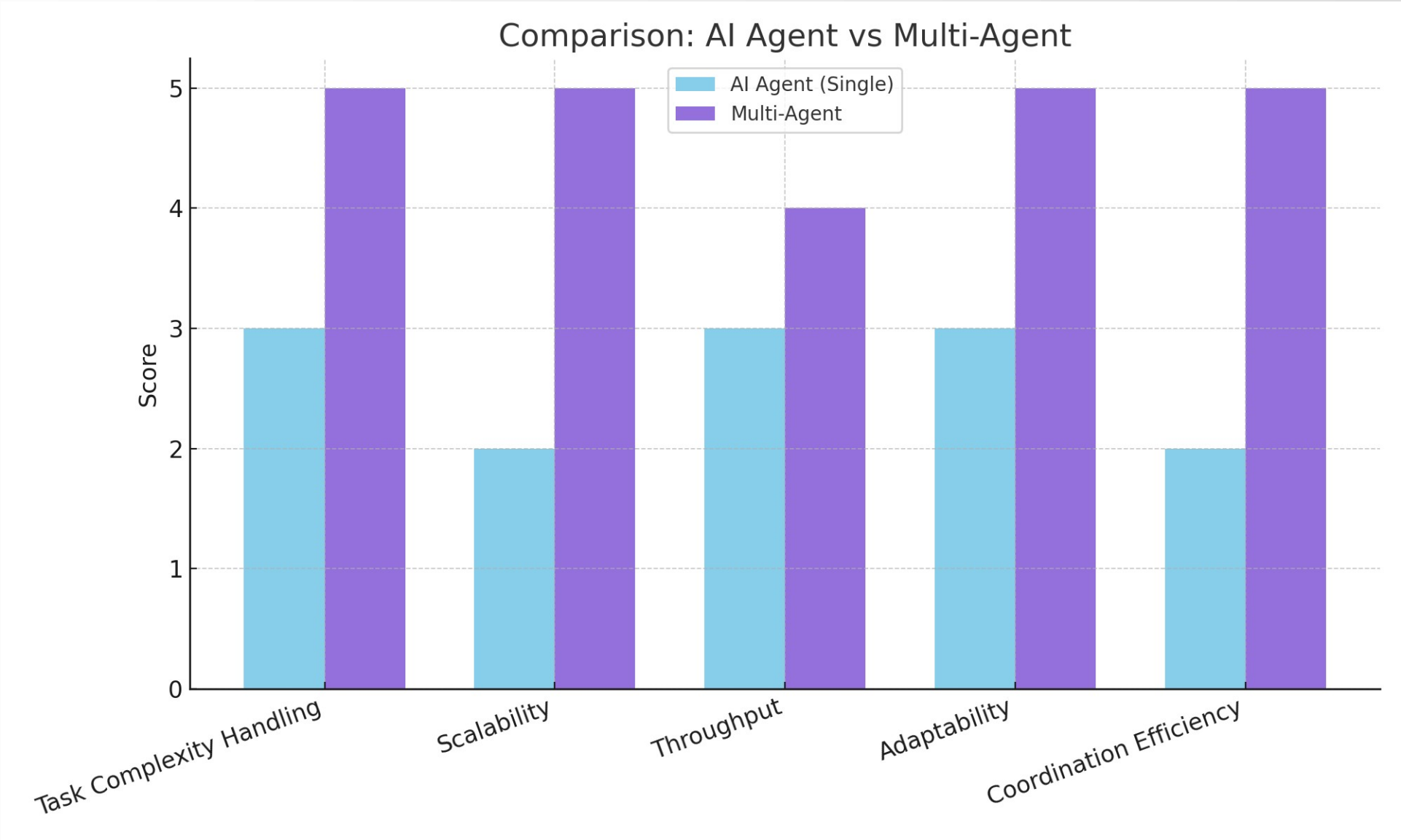
- One decision maker
- Linear execution
- Fixed environment
- Isolated problem-solving

Multi-Agent System

- Multiple interacting agents
- Dynamic collaboration
- Parallel execution
- Distributed decision-making



AI Agent vs Multi-Agent





Multi-Agent 的流量特征

- 突发性：业务场景的特殊性，导致流量会出现集中和突发
- 异构性：多种类型的流量，文本 / 图像 / 视频 / 音频等
- 会话相关性：上下文需要保持
- 延迟敏感性：一些实时交互场景对延迟敏感



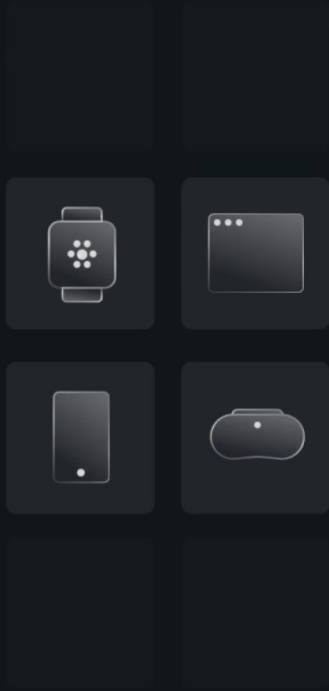
传统网关的不足

- 缺乏对 Token 的第一类支持
- 负载均衡策略过于简单
- 缺乏对 AI workload 的感知
- 静态配置适应性不足

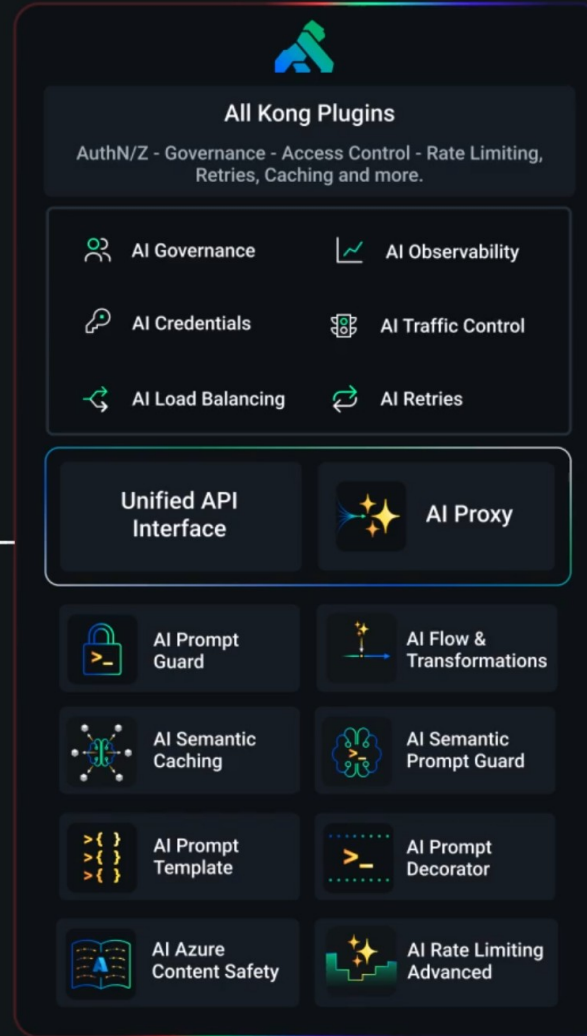


Kong AI Gateway

Apps & Agents

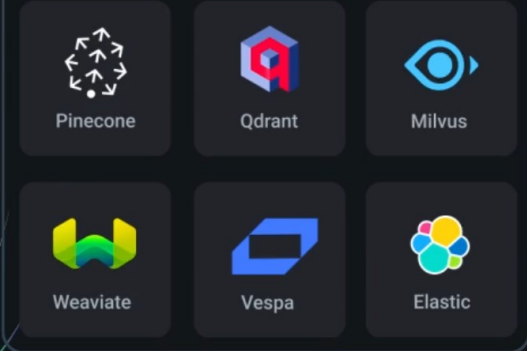


AI Gateway

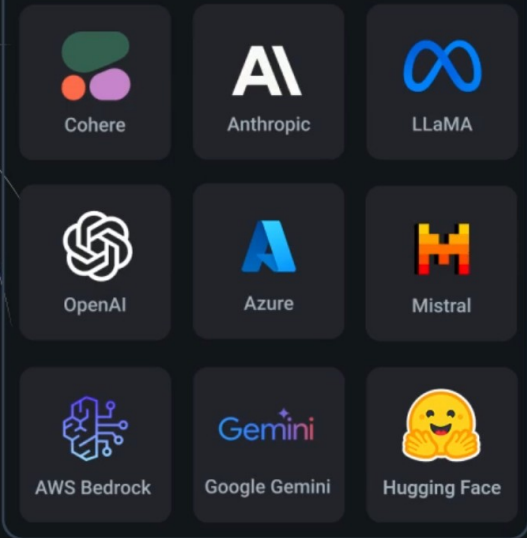


AI Providers

Vector DBs



LLMs

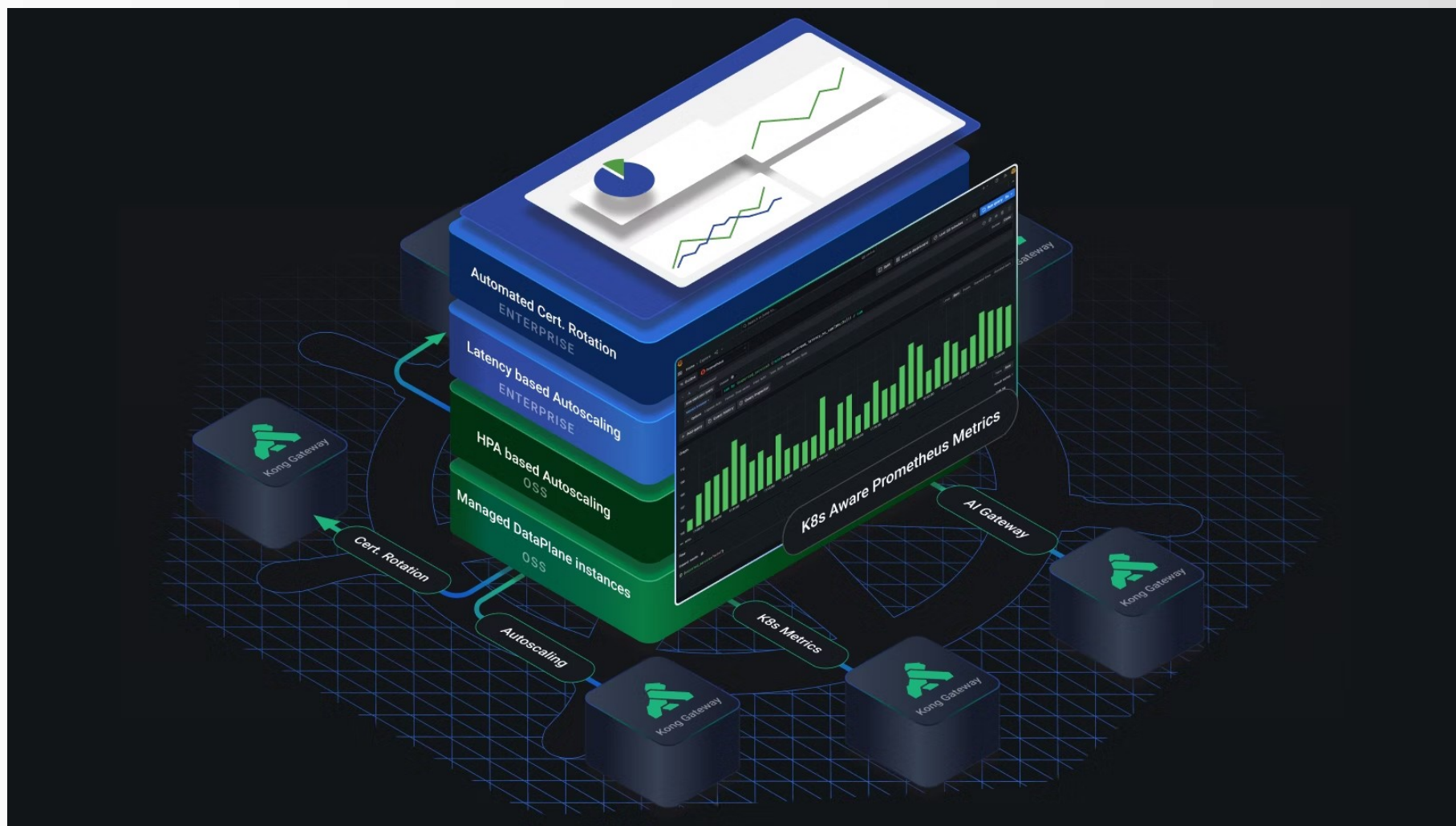


Kong **Konn**ect





Kong Gateway Operator





Kong Gateway Operator



KONECT

Overview

Gateway Manager

AI Manager

Overview

Load balancing

Prompt management

Guardrails

RAG

PII sanitization

Semantic caching

Go to LLM analytics

Mesh Manager

API Products

Dev Portal

Analytics

Service Catalog

MoeLove

AU (Australia)

Quick search...

Ctrl K

?

?

?

JZ

Mistral, Cohere

Created: 5/28/2025

ID: 544bf54c-5de5-43f9-b8df-05c46c54d084

Identified consumers: 1 agent

Models: Mistral/* Cohere/*

Manage LLM

Control plane: moelove

Load balancing: round-robin (2 models)

LLM authentication: Provided by client and AI gateway

Gateway service: AIManagerModelService_1748370203280

Streaming: Enabled

Tiers of access: One for all clients

An LLM model is a Gateway Service in your control plane whose access you can manage like any other services or APIs. Unlike an API though, you can apply AI plugins to an LLM to govern usage, security and more.

LLM Analytics

Requests: 12

Tokens: 1040

Error Rate: 66.67%

LLM Average Latency: 2338.19ms

100% vs previous 30 days

100% vs previous 30 days

66.67% vs previous 30 days

100% vs previous 30 days

Refreshing...

Last 30 days consumption

Anonymous

mistral

UNSPECIFIED

cohere

7 requests - 2376ms avg latency

3 requests - 0ms avg latency

2 requests - 1403ms avg latency



Kong Gateway Operator

- 全声明式配置
- 35+ CRDs
- 分拆核心 / 扩展 / 配置
- 证书自动轮转
- 多纬度的流量控制和感知
- 缓存控制
- 性能优化

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization

resources:
- configuration.konghq.com_kongcacertificates.yaml
- configuration.konghq.com_kongcertificates.yaml
- configuration.konghq.com_kongconsumergroups.yaml
- configuration.konghq.com_kongconsumers.yaml
- configuration.konghq.com_kongcredentialaccls.yaml
- configuration.konghq.com_kongcredentialapikeys.yaml
- configuration.konghq.com_kongcredentialbasicauths.yaml
- configuration.konghq.com_kongcredentialhmacs.yaml
- configuration.konghq.com_kongcredentialjwts.yaml
- configuration.konghq.com_kongdataplaneclientcertificates.yaml
- configuration.konghq.com_kongkeys.yaml
- configuration.konghq.com_kongkeysets.yaml
- configuration.konghq.com_konglicenses.yaml
- configuration.konghq.com_kongpluginbindings.yaml
- configuration.konghq.com_kongplugins.yaml
- configuration.konghq.com_kongroutes.yaml
- configuration.konghq.com_kongservices.yaml
- configuration.konghq.com_kongsnis.yaml
- configuration.konghq.com_kongtargets.yaml
- configuration.konghq.com_kongupstreams.yaml
- configuration.konghq.com_kongvaults.yaml
- gateway-operator.konghq.com_aigateways.yaml
- gateway-operator.konghq.com_controlplanes.yaml
- gateway-operator.konghq.com_dataplanemetricsextensions.yaml
- gateway-operator.konghq.com_dataplanes.yaml
- gateway-operator.konghq.com_gatewayconfigurations.yaml
- gateway-operator.konghq.com_kongplugininstallations.yaml
- gateway-operator.konghq.com_konnectextensions.yaml
- gateway-operator.konghq.com_watchnamespacegrants.yaml
- konnect.konghq.com_konnectapiauthconfigurations.yaml
- konnect.konghq.com_konnectcloudgatewaydataplanegroupconfigurations.yaml
- konnect.konghq.com_konnectcloudgatewaynetworks.yaml
- konnect.konghq.com_konnectcloudgatewaytransitgateways.yaml
- konnect.konghq.com_konnectextensions.yaml
- konnect.konghq.com_konnectgatewaycontrolplanes.yaml
```



```
_format_version: "3.0"

services:
  - name: ai
    url: https://localhost:32000

  routes:
    - name: openai-gpt4o
      paths:
        - "/gpt4o"
      plugins:
        - name: ai-proxy
          config:
            route_type: llm/v1/chat
            model:
              provider: openai
              name: gpt-4o
            auth:
              header_name: Authorization
              header_value: "Bearer <OPENAI_KEY_HERE>" # replace with your OpenAI key again

# Now we add a security plugin at the "individual model" scope
  - name: key-auth
    config:
      key_names:
        - Authorization

# and finally a consumer with **its own API key**
consumers:
  - username: department-1
    keyauth_credentials:
      - key: "Bearer department-1-api-key"
```




Multi-Agent 的适配增强

- Token-aware 限流策略
- Semantic caching
- 智能负载均衡
- Agent 身份识别与认证
- 动态路由策略
- 会话亲和性管理
- MCP (Model Context Protocol) 集成
- Auto RAG
- ...



Token-aware

Tokens

7

Characters

23

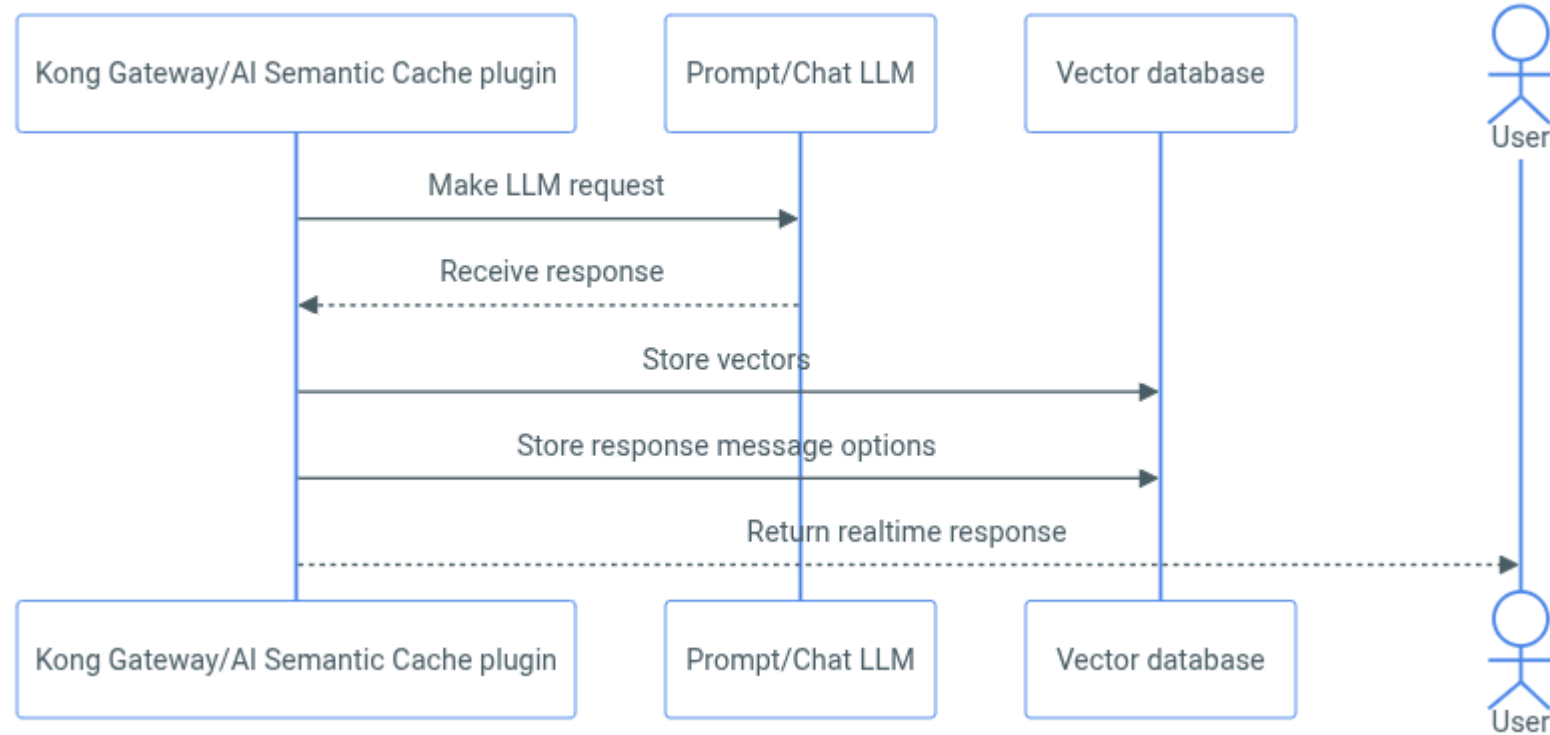
Kong AI Gateway at GIAC

Text

Token IDs



Semantic caching





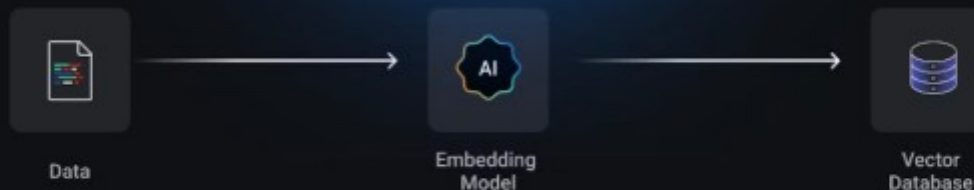
```
{
  "name": "ai-semantic-cache",
  "config": {
    "embeddings": {
      "auth": {
        "header_name": "Authorization",
        "header_value": "Bearer OPENAI_TOKEN"
      },
      "model": {
        "provider": "openai",
        "name": "text-embedding-3-large"
      }
    },
    "vectordb": {
      "redis": {
        "host": "REDIS_HOST",
        "port": 6379
      },
      "strategy": "redis",
      "dimensions": 768,
      "distance_metric": "cosine",
      "threshold": 0.25
    },
    "cache_ttl": 300,
    "cache_control": true,
    "exact_caching": true
  }
}
```



Multi-Agent 的适配增强

Automated RAG with Kong

1. Processing Data



2. Associating relevant data to prompt



- ✓ Lower LLM hallucinations
- ✓ Improve dev productivity
- ✓ Grow secure



下一步展望

- Kong Gateway Operator => Kong Operator
- 更智能的调度算法
- 标准化推进
- 开源社区贡献

GIAC

全球互联网架构大会

GLOBAL INTERNET ARCHITECTURE CONFERENCE

连接未来的技术架构

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